2.3 The Derivative of a Polynomial Function; The Derivative of $y = e^x$

Name _____ Period ____ Date _____

AP Calculus 2.3 · Student sheet

I can…

- use the Constant, Power, Sum, and Constant Multiple Rules.
- find the derivative of $y = e^x$.
- use derivatives to find tangent lines, horizontal tangents, and velocity.

Vocabulary · fill in the missing word

Leibniz notation — $\frac{dy}{dx}$ = _____ = $f'(x)$

Simple Power Rule — $\frac{d}{dx} x^n$ = _____

Derivative of e^x — $\frac{d}{dx} e^x$ = _____

Part 1 · Find the pattern

Look for the pattern.

f(x)	x^2	x^3	x^4	x^{10}
$f'(x)$	$2x$	$3x^2$		

Part 2 · Notes and examples (fill in each note, then work the example)

Constant and Power Rules: $d/dx\ c = 0$ and $d/dx\ x^n$ = _____.

Example 1

Find each derivative.

a. $\frac{d}{dx} x^5$

b. $\frac{d}{dx} 7$

c. $\frac{d}{dx} x$

Sum and Constant Multiple Rules: $d/dx\ [kf(x) + g(x)] = kf'(x) +$ _____.

Example 2

Find $f'(x)$ for $f(x) = 3x^4 - 2x^2 + x - 5$.

Example 3

Find the tangent line to $f(x) = x^3 + 3x - 1$ at $(0, -1)$.

The exponential function: $\frac{d}{dx} e^x =$ _____.

Example 4

Find $f'(x)$ and $f'(0)$ for $f(x) = 4e^x + x^3$.

Example 5

Where does $f(x) = x^3 - 3x + 2$ have a horizontal tangent line?

Part 3 · Partner practice

Partner 1

$s(t) = t^3 - 6t^2 + 9t$. Find the velocity at $t = 2$.

Partner 2

(AP® Practice) If $f(x) = e^x$, then $\ln(f'(3)) =$

(A) 3 (B) 0 (C) e^3 (D) $\ln 3$